

Algebra 1
Unit 9
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

Kamil's work when solving the equation $x^2 = -3x + 40$ using the quadratic formula is shown. Use Kamil's work to answer questions 1 and 2.

$$x^2 = -3x + 40, \text{ so } a = 1, b = -3, c = 40$$

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(1)(40)}}{2(1)}$$

$$x = \frac{3 \pm \sqrt{9 - 160}}{2}$$

$$x = \frac{3 \pm \sqrt{-151}}{2}$$

Two Non-real Solutions

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(-1)(40)}}{2(-1)}$$

$$x = \frac{3 \pm \sqrt{9 + 160}}{-2}$$

$$x = \frac{3 \pm \sqrt{169}}{-2}$$

$$x = -8, 5$$

$$x = \frac{3 \pm 13}{-2}$$

$$x = \frac{3 + 13}{-2} = \frac{16}{-2} = -8$$

$$x = \frac{3 - 13}{-2} = \frac{-10}{-2} = 5$$

- Circle the step where Kamil made an error.
- Correct Kamil's error and the steps that follow to find the correct solution(s) to the equation. Use the space provided above to show the correct steps.

Solve each equation by using the quadratic formula. Show your work.

3. $2x^2 - x - 6 = 0$

$$x = \frac{1 \pm \sqrt{(-1)^2 - 4(2)(-6)}}{2(2)}$$

$$x = \frac{1 \pm \sqrt{49}}{4}$$

$$x = \frac{1 \pm 7}{4}$$

$$x = \frac{1 + 7}{4} = \frac{8}{4} = 2$$

$$x = \frac{1 - 7}{4} = \frac{-6}{4} = -\frac{3}{2}$$

4. $3x^2 + 3 = -6x$

$$3x^2 + 6x + 3 = 0$$

$$x = \frac{-6 \pm \sqrt{(6)^2 - 4(3)(3)}}{2(3)}$$

$$x = \frac{-6 \pm \sqrt{36 - 36}}{6}$$

$$x = \frac{-6 \pm \sqrt{0}}{6}$$

$$x = \frac{-6 \pm 0}{6} = -1$$

$$5. \quad 0 = x^2 + 2x - 8$$

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-8)}}{2(1)}$$

$$x = \frac{-2 \pm \sqrt{4 + 32}}{2}$$

$$x = \frac{-2 \pm 6}{2}$$

$$x = \frac{-2 + 6}{2} = \frac{4}{2} = 2$$

$$x = \frac{-2 - 6}{2} = \frac{-8}{2} = -4$$

$$7. \quad 8x^2 - 4x - 10 = 8$$

$$8x^2 - 4x - 18 = 0$$

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(8)(-18)}}{2(8)}$$

$$x = \frac{4 \pm \sqrt{592}}{16}$$

$$x = \frac{4 \pm \sqrt{592}}{16}$$

$$x = \frac{4 \pm 4\sqrt{37}}{16}$$

$$x = \frac{1 \pm \sqrt{37}}{4}$$

$$9. \quad 10x^2 - 7x = 2x^2 - 1$$

$$8x^2 - 7x + 1 = 0$$

$$x = \frac{7 \pm \sqrt{(-7)^2 - 4(8)(1)}}{2(8)}$$

$$x = \frac{7 \pm \sqrt{49 - 32}}{16}$$

$$x = \frac{7 \pm \sqrt{17}}{16}$$

$$6. \quad 2x^2 - 2x + 3 = 3x$$

$$2x^2 - 5x + 3 = 0$$

$$x = \frac{5 \pm \sqrt{(-5)^2 - 4(2)(3)}}{2(2)}$$

$$x = \frac{5 \pm \sqrt{1}}{4}$$

$$x = \frac{5 + 1}{4} = \frac{6}{4} = \frac{3}{2}$$

$$x = \frac{5 - 1}{4} = \frac{4}{4} = 1$$

$$8. \quad -9x = x^2 + 20$$

$$x^2 + 9x + 20 = 0$$

$$x = \frac{-9 \pm \sqrt{(9)^2 - 4(1)(20)}}{2(1)}$$

$$x = \frac{-9 \pm \sqrt{81 - 80}}{2}$$

$$x = \frac{-9 \pm 1}{2}$$

$$x = \frac{-9 + 1}{2} = \frac{-8}{2} = -4$$

$$x = \frac{-9 - 1}{2} = \frac{-10}{2} = -5$$

$$10. \quad -4x - 14 = -x^2$$

$$x^2 - 4x - 14 = 0$$

$$x = \frac{4 \pm \sqrt{(4)^2 - 4(1)(-14)}}{2(1)}$$

$$x = \frac{4 \pm \sqrt{16 + 56}}{2}$$

$$x = \frac{4 \pm \sqrt{72}}{2}$$

$$x = \frac{4 \pm 6\sqrt{2}}{2}$$

$$x = 2 \pm 3\sqrt{2}$$